

E-Commerce Analytics Dashboard – Business Insights Report

Project Overview

This project analyzes e-commerce sales data to understand business performance, customer behavior, product performance, and regional sales trends. The objective is to transform raw transactional data into actionable business insights using Python, Pandas, and Power BI.

Key Performance Indicators (KPIs)

- Total Sales: \$2.30M
- Total Profit: \$286.40K
- Total Orders: 5,009
- Total Customers: 793

These KPIs provide a high-level overview of the company's overall business performance.

Sales Analysis

Category Performance

- Technology generated the highest revenue among all categories.
- Office Supplies maintained consistent sales across regions.
- Furniture contributed significant sales but lower profit margins.

Product Performance

- The top 10 products contributed a major portion of total revenue.
 - Certain products showed high sales volume but relatively low profitability.
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Regional Analysis

Best Performing Region

- The West region generated the highest sales and profit.
- East region also contributed significantly to overall revenue.

Low Performing Region

- South region generated comparatively lower sales.
 - Additional marketing campaigns may improve performance in this region.
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Customer Analysis

Top Customers

- A small percentage of customers contributed a significant share of total revenue.
- High-value customers should be targeted with loyalty programs and personalized offers.

Customer Segments

- Consumer segment generated the largest portion of sales.
 - Corporate and Home Office segments provide opportunities for growth.
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Sales Trend Analysis

- Sales exhibited seasonal patterns throughout the year.
 - Revenue increased significantly during peak shopping months.
 - Inventory planning should account for seasonal demand fluctuations.
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Profitability Analysis

- Excessive discounts negatively impacted profit margins.
 - Products with high discounts should be reviewed to ensure profitability.
 - Pricing optimization can improve overall profit performance.
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Business Recommendations

Revenue Growth

- Focus marketing efforts on top-performing categories.
- Increase inventory for high-demand products.

Customer Retention

- Introduce customer loyalty programs.
- Offer personalized promotions for repeat customers.

Regional Expansion

- Strengthen marketing in underperforming regions.
- Investigate operational factors affecting regional performance.

Profit Optimization

- Reduce unnecessary discounts.
 - Prioritize products with strong profit margins.
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Conclusion

The analysis reveals strong overall business performance with significant opportunities for improving profitability, customer retention, and regional expansion. Data-driven decision-making can help the organization maximize revenue and sustain long-term growth.